



Mapping complexity of spatial distribution of faults using fractal and multifractal models: vectoring towards exploration targets

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ABSTRACT

In this paper, fractal and multifractal analyses are demonstrated as effective tools for mapping complexity in the spatial distribution of faults. Faults within the eastern part of Gejiu mining area, Yunnan province, west southern China were chosen to demonstrate mapping of the complexity of their spatial distributions using fractal and multifractal models. The results show that (1) the fractal dimensions of the spatial distributions of all faults, NW-trending faults, and NE-trending faults are 1.68, 1.49, and 1.42, respectively, indicating differences in spatial distributions of different sets of faults; (2) the fractal dimensions of the spatial distributions of faults in the four Sn fields in the Gejiu mining district, namely Malage, Gaosong, Laochang, and Kafang (arranged in the order of increasing proportions of surface-projected areas of Sn orebodies) are 1.38, 1.57, 1.65, and 1.41, respectively; and (3) complexity of the spatial distributions of faults, represented by fractal dimension, correlates well with surface-projected areas of Sn orebodies, and lengths of faults satisfy the multifractal statistical and singularity index α , which can be used to quantify the complexity of the spatial distributions of faults.

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1. Introduction

There is increasing interest in applying fractal geometry in quantitative descriptions of natural phenomena, such as spatial distribution of mineral deposits. Fractal and multifractal analyses have been demonstrated as useful tools for identifying irregularity in patterns of natural objects (Mandelbrot, 1983). In the geosciences, fractal dimension and multifractal spectra have increasingly been applied to describe complexity and self-similarity of geological objects, geological processes, and geological events (Carlson, 1991; Cheng et al., 1994, 1996; Zuo et al., 2009a, 2009c; Gumiel et al., 2010). Fractal analyses have also been used to infer fault controls on mineralization, which are important in predictive mapping of prospectivity for certain types of mineral deposits (Carranza, 2008, 2009, 2011; Carranza et al., 2009; Carranza and Sadeghi, 2010; Zuo et al., 2009b; Zuo, 2011).

Fault systems provide pathways for focusing large volumes of mineralizing fluids required to concentrate and form mineral deposits. Since fault arrays were recognized to have fractal geometry (Tchalenko, 1970), subsequent works have demonstrated that

fractal properties of faults are key to understanding fault formation, growth, and strain. Some researchers have described spatial patterns and spatial distributions of faults in terms of fractal geometry and several analyses have been carried out showing the relevance of fractal dimension (Hirata, 1989; Velde et al., 1990; Walsh and Watterson, 1993; Pérez-López and Paredes, 2006). Cowie et al. (1995) found that fault populations exhibit multifractal scaling properties. Agterberg et al. (1996) demonstrated that surface fractures can be modeled as multifractals. Fractal dimensions and multifractal spectra represent synthetic characterizations of fault scale, evolution, assembly pattern, dynamics, and mechanics, which act as effective and accurate quantitative indices of the complexity in spatial distribution of various fault sets (e.g., Korvin, 1992; Walsh and Watterson, 1993; Kruhl, 1994; Agterberg et al., 1996; Scholz, 1997; Koukouvelas et al., 1999; Pérez-López et al., 2005).

High fractal dimensions of spatial patterns of faults/fractures are thought to be beneficial for fluid flow, fluid gathering, and mineral or hydrocarbon deposition (Barton and La Pointe, 1995; Xie and Tan, 2002). The fractal dimension of the spatial distribution of fault sets has been used in various ways to study fault systems. Yao and Zhan (2007) used fractal geometry to study the relationship between the spatial distribution of fault systems and the oil distribution in the southern part of the South China Sea. Hodkiewicz et al. (2005) used fractal geometry to quantify complexity in map patterns of structures and lithological

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contacts. Ford and Blenkinsop (2008) used fractal geometry to evaluate the relationship between complexity, complexity gradients, and copper mineralization in Mt. Isa Inlier.

In this paper, fractal and multifractal methods are described for characterizing the complexity of spatial distributions of faults in two-dimensional space, and to determine whether or not there exists a relationship between the complexity of the spatial distribution of faults and the mineral resource.

2. Geological setting of the Gejiu district

The Gejiu district is situated along the suture zone between the Indian and the Eurasian plates at the southwestern edge of the

southwestern China subplate. The district is located ca. 200 km south of the city of Kunming, the capital of Yunnan Province (Fig. 1). The district contains world-class Sn and Cu deposits plus abundant deposits of nonferrous and rare minerals rich in Pb, Zn, W, Ag, Bi, and In (Zhuang et al., 1996).

The district consists of two parts, namely the eastern and western parts divided by the Gejiu fault (Fig. 1). This study focuses on the eastern part, which contains most of the Sn resources in the district. From north to south, the eastern part of the study area can be divided into four Sn fields: Malage, Gaosong, Laochang, and Kafang (Fig. 2).

The Gejiu district has undergone a prolonged history of tectonic and complex structural processes that resulted in various scales of structural systems. The Wuzhishan anticlinorium is the main host of Sn deposits in the study area (Fig. 1). Faults are extremely well developed in the study area and are mainly dominated by NW trending and NE trending sets (Fig. 2). These sets of faults are closely related to deposit-forming processes in the study area. They acted as pathways of traps of mineralization fluid, and, thus, controlled the occurrences and spatial distributions of Sn and other mineral deposits in the study area (Zhuang et al., 1996; Ma et al., 2004). The main types of fault-controlled Sn deposits within the four Sn fields include interbedded vein-type deposits (Type I), interbedded stratiform and banded oxidized deposits (Type II), skarn-type deposits (Type III), and stockwork of tourmaline-type deposits (Type IV) (Zhuang et al., 1996; Zhang et al., 2008). Each Sn fields contains at least two or more types of Sn deposits (Table 1).

The Sn resources in the Laochang, Gaosong, Kafang, and Malage fields account for, respectively, 40%, 35%, 15%, and 10% of the total Sn resources in the Gejiu district (Zhuang et al., 1996; Zhang et al., 2008). The Sn grades in the district range from 0.5% to 1.2%, and all the Sn fields have similar average Sn grades. There exists a positive relationship between the district-wise proportions of Sn resources and the surface-projected areas of Sn orebodies in the four Sn fields (Table 1).

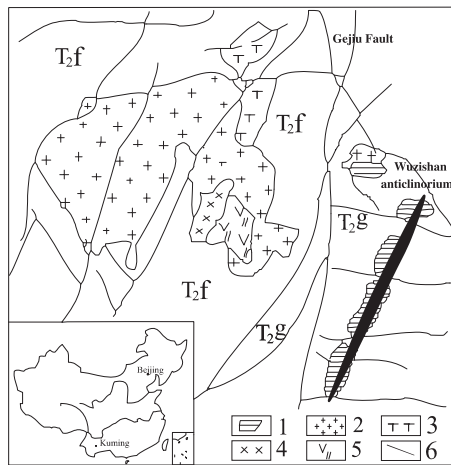


Fig. 1. Simplified geological map of the Gejiu district (modified from Zhang et al., 2007): 1. main ore fields; 2. granite; 3. alkalic rocks; 4. gabbro; 5. monzonite; and 6. faults.

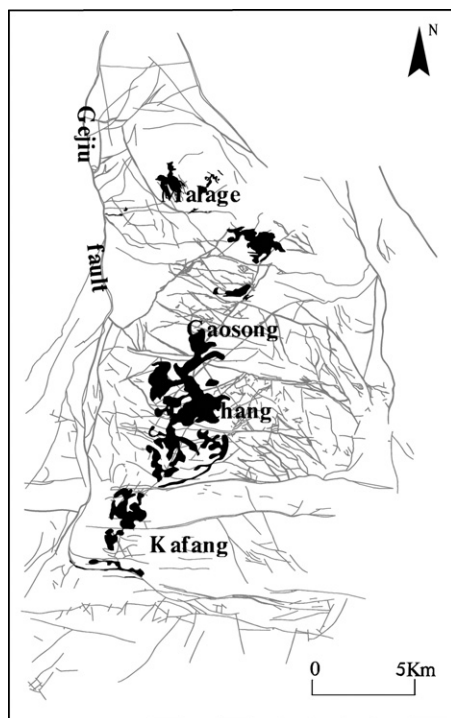


Fig. 2. Mapped faults (gray polylines) and surface projections (polygons in black) of Sn orebodies at depth in the eastern part of the Gejiu district.

3. Methodology

3.1. Fractal and multifractal models

Fractal and multifractal models are effective tools for characterizing complex physical processes and their end-products (Cheng, 2000, 2003), as observed from many end products of singular processes that can be modeled by power-law relations describing scale invariance and self-similarity properties. The fractal dimension can be obtained by the box-counting method (Velde et al., 1990; Walsh and Watterson, 1993; Borodich, 1997; Jiang, 2005; Pérez-López et al., 2005; Zuo et al., 2009a, 2009c). The box-counting method is among the number-size fractal models that have been used for fault analysis. In a log-log graph, the box-counting method shows a power-law relationship between the cumulative number and the cumulative size of similarly shaped objects. If all faults in a given area are covered in a raster map with different cell sizes, this relationship can be expressed as

$$N(r) \propto kr^{-D}, \quad (1)$$

where r is a measure of unit size, $N(r)$ is cumulative number of cells containing faults, D is box-counting fractal dimension, $\propto$ means “proportional to”, and k is a constant. The D is obtained as the slope coefficient of linear regression between values of $N(r)$ and r . The value of D varies between 1 and 2 for a two-dimensional map (Mandelbrot, 1983).

Fractal sets can be measured by determining their presence or absence in collections of cells created by partitioning of

Table 1

Sn fields in the Gejiu district: Sn deposit attributes, fault attributes, and data/results of box-counting fractal analysis of fault distributions.

Sn field	Sn deposit attributes			Fault attributes		$N(r)$					D	R^2	SE
	Types	Surface-projected areas (km ²) of orebodies	% of resources in the district	No. of segments	Total length (km)	$r=0.2$ km	$r=0.4$ km	$r=0.5$ km	$r=0.8$ km	$r=1.0$ km			
Malage	I, III	1.4	10	78	128.24	663	274	198	100	72	1.38	0.99	0.011
Gaosong	II, III	5.2	35	230	268.63	718	258	187	83	57	1.58	0.99	0.014
Laochang	III, IV	9.6	40	284	334.55	1155	425	285	120	83	1.65	0.99	0.019
Kafang	III, IV	3.1	15	86	143.66	568	229	168	83	58	1.42	0.99	0.014

Abbreviations: $N(r)$, cumulative number of square cells, with side r , containing faults (see Eq. (1)); D , box-counting fractal dimension; R^2 , coefficient of determination explained by regression; SE, standard error of regression.

k -dimensional space R ($k=1, 2$, or 3) (Cheng et al., 1994). For multifractals, the amount of random variable is measured in the collections of cells. To quantify the complexity of the spatial distribution of faults, mapped faults are converted into a raster with a spatial resolution of ε km $\times$ ε km cell. Let $\mu(\varepsilon)$ represent the total length of faults within each cell. The partition function can be then defined as

$$\chi_q(\varepsilon) = \sum_{N(\varepsilon)} \mu_i^q(\varepsilon), \quad (2)$$

where $N(\varepsilon)$ is total number of unit cells of size ε . If $\mu(\varepsilon)$ satisfies the multifractal model, the partition function $\chi_q(\varepsilon)$ and the size ε satisfy a power-law relationship

$$\chi_q(\varepsilon) = \varepsilon^{\tau_q}, \quad (3)$$

where τ_q is mass exponent of order q . Using τ_q , singularity index α and multifractal spectrum $f(\alpha)$ can be calculated as (Evertsz and Mandelbrot, 1992; Cheng, 1997, 1999), respectively,

$$\alpha(q) = \frac{\partial \tau(q)}{\partial q} \quad (4)$$

and

$$f(\alpha) = \alpha(q) - \tau(q), \quad (5)$$

where $\alpha(q)$ is a decreasing function of q and $f(\alpha)$ is a convex function of α . The ranges of these two functions and the curvature of function $\tau(q)$ indicate the degree of multifractality (Cheng, 2002). More details about the properties of the multifractal spectrum for earth sciences applications can be found in Falconer (1990), Cheng and Agterberg (1995), Agterberg et al. (1996), Cheng (1997a, 1997b), and Panahi and Cheng (2004).

3.2. Procedure for fractal and multifractal analyses in this study

The mapped faults in the study area (Fig. 2) are based on the Yunnan Tin Group maps of 1:50,000 scale. The map of faults used (Fig. 2) covers ca. 600 km². It has a roughly uniform level of mapping detail in the individual Sn fields, and thus it was considered appropriate for this fractal study. The fractal analysis was carried out separately for the map of all faults and for maps of NW- and NE-trending faults as follows:

- (1) Each fault map is converted into separate raster maps with spatial resolutions (r) of 400, 500, 800, 1000, 2000, and 4000 m. In each raster map, the number of cells $N(r)$ occupied by faults is counted. Then, a log–log plot of $N(r)$ versus r for each fault map is constructed.
- (2) Each map of point locations of fault-controlled Sn deposits in each of the four Sn fields (i.e., Malage, Gaosong, Laochang, and Kafang) is converted into separate raster maps with spatial

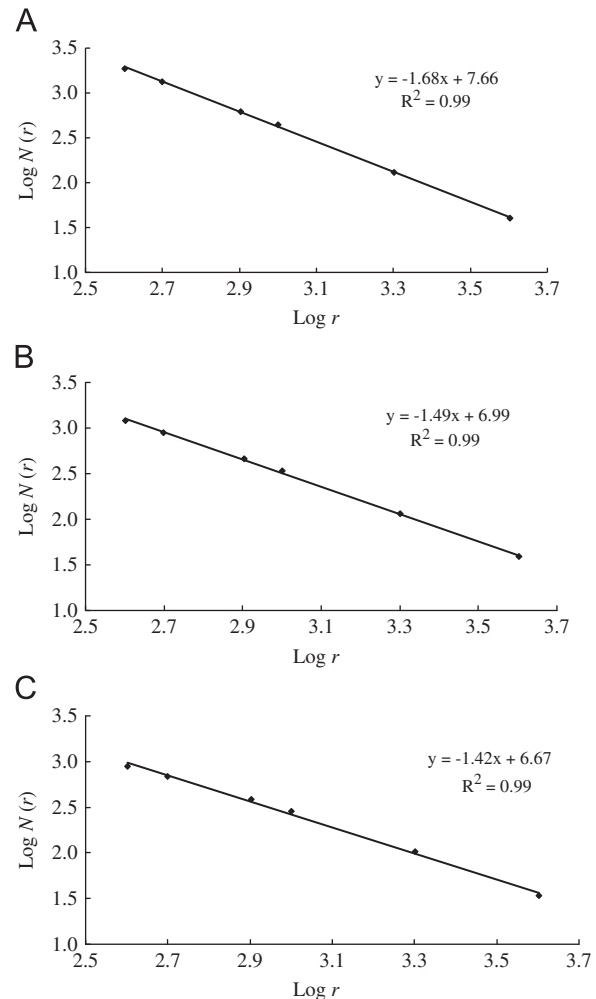


Fig. 3. Log–log plots (base 10) of cell size r versus the number of cells $N(r)$ occupied by (A) all faults, (B) NW-trending faults, and (C) NE-trending faults.

resolutions (r) of 200, 400, 500, 800, and 1000 m. Then, the box-counting analysis described in step (1) is performed.

- (3) Each fault map is divided into square cells of 1×1 km². Fractal dimension D in every cell in the whole study area is calculated using the box-counting method for spatial resolutions (r) of 1000, 500, 250, and 125 m. The value of D derived for each cell is attributed to its center. A contour map of fractal dimensions is obtained by interpolation of the D values.
- (4) Each fault map is converted into separate raster maps with different spatial resolutions ε , and the partition function $\chi_q(\varepsilon)$

Table 2

Pearson's correlation coefficients among Sn deposit attributes, fault attributes, and fractal dimensions of fault distributions in the four Sn fields in the Gejiu district (see Table 1).

	D	FN	FL	RA	RP
D	1.00				
FN	1.00	1.00			
FL	1.00	1.00	1.00		
RA	0.95	0.93	0.96	1.00	
RP	1.00	0.99	0.99	0.92	1.00

Abbreviations: D, fractal dimension; FN, no. of fault segments; FL, total length of faults; RA, surface-projected areas of Sn orebodies; RP, % of Sn resources in the district.

is used to build the multifractal model for the total length of faults in each grid. The component $\tau(q)$, singularity index α , and multifractal spectrum $f(\alpha)$ are used to illustrate the multifractal characteristics of each fault set.

4. Results and discussion

4.1. Fractal dimensions of fault patterns

The value of a fractal dimension may be obtained as the slope of a straight line fitted to the log–log plot of r versus $N(r)$ (Fig. 3), to represent the box-counting dimension (D) of a set of faults

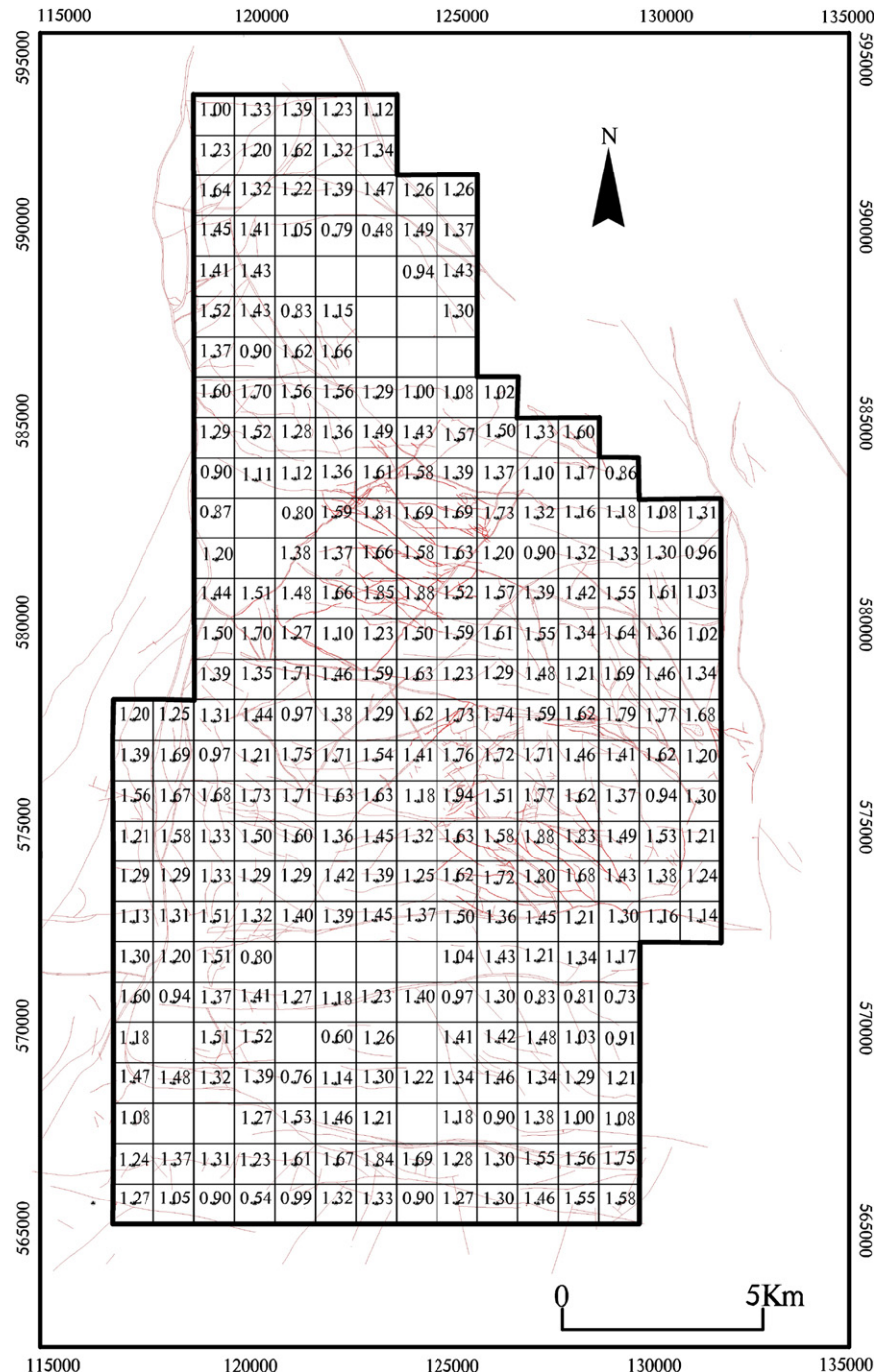


Fig. 4. Estimates of fractal dimensions of the spatial distributions of faults in every $1 \times 1 \text{ km}^2$ cell in the study area.

(Table 1). The value of D for the map of all faults was 1.68 ± 0.010 , for the map of NE-trending faults 1.42 ± 0.014 , and for the map of NW-trending faults 1.49 ± 0.013 . The values of D show that the spatial distributions of faults in each of the four Sn fields are self-similar or invariant within the scale interval of 200–1000 m (Table 1). The fractal dimensions of the spatial distributions of faults in the four Sn fields decrease in the following order: Laochang (1.65 ± 0.019) > Gaosong (1.58 ± 0.014) > Kafang (1.41 ± 0.014) > Malage (1.38 ± 0.011). In addition, the numbers and lengths of faults in the four Sn fields follow the same decreasing trend of D values in the four Sn fields.

4.2. Complexity of spatial distribution of faults

Hydrothermal mineral systems develop in active faulted parts of the earth's crust where and when there is sufficient connectivity of fluid pathways to favor formation of mineral deposits (Cox et al., 2001). In order to form hydrothermal mineral deposits, the flow of hydrothermal fluids generated at depth in the earth's crust must be focused toward high-permeability zones where they are trapped and eventually precipitate metals under certain favorable temperature–pressure conditions. High degrees of interconnectivity of faults are, therefore, essential for

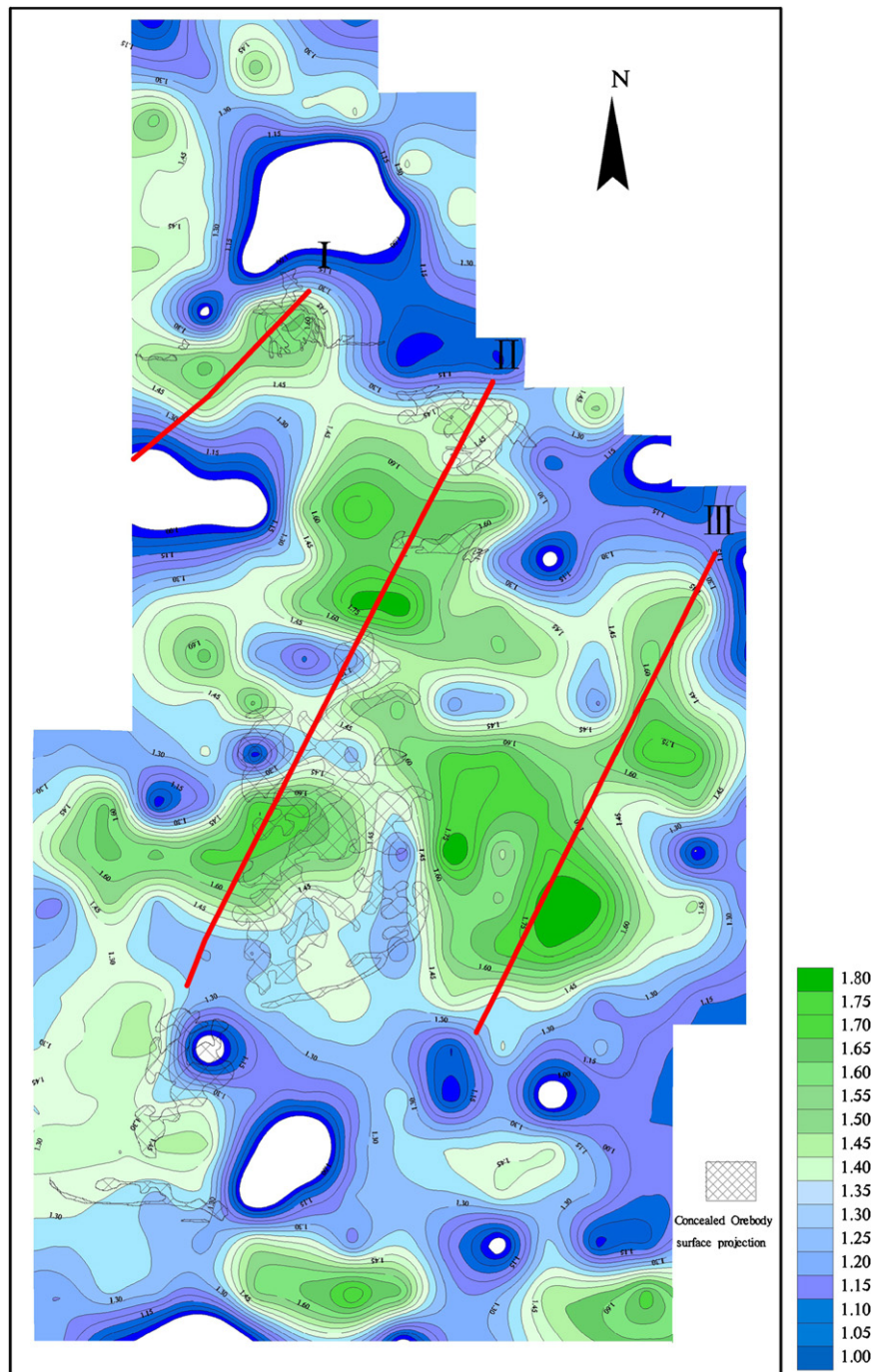


Fig. 5. Contour map of interpolated estimates of fractal dimensions of fault spatial distributions per $1 \times 1 \text{ km}^2$ cell (see Fig. 4).

hydrothermal mineralization. It has been shown that regions with large fractal dimensions represent zones with a high probability of having well-connected faults of fluid pathways (Hodkiewicz et al., 2005). Therefore, fractal dimension is an index of complexity of faults or interconnectivity of fluid pathways.

The derived fractal dimensions of spatial distributions of all faults in each of the Sn fields have strong positive correlations with numbers of fault segments and lengths of faults (Table 2). In the Sn fields, surface-projected areas of Sn orebodies and percentage of Sn resources in the district also have strong positive correlations with the numbers of fault segments and lengths of faults. The correlations shown in Table 2 reflect that areas of high complexity (i.e., with high fractal dimensions) of the spatial distribution of faults have the potential to host large mineral resources.

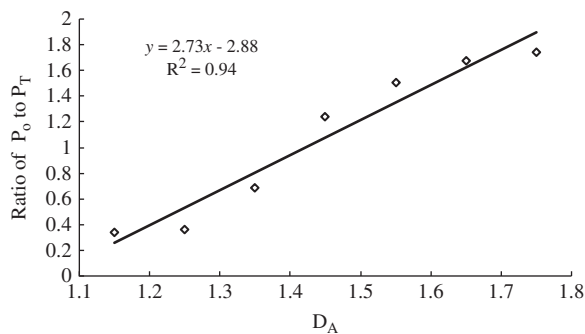


Fig. 6. Plot of ratios of surface-projected areas of individual orebodies (P_o) to total surface-projected areas of orebodies area (P_T) versus average fractal dimension (D_A) surface-projected areas of individual orebodies obtained from Figs. 2 and 5.

To describe further the spatial distribution of faults in the Gejiu district, fractal dimensions of faults patterns were calculated in $1 \times 1 \text{ km}^2$ grid cells in the whole study area using the box-counting method. The grid cell size chosen is related to the map scale, and smaller grid cells must be applied to maps with higher scales. The $1 \times 1 \text{ km}^2$ grid cell size was chosen based on the 1:50,000 scale of the fault map. The box sizes used for measuring the fractal dimension in every $1 \times 1 \text{ km}^2$ cell were 1, 0.5, 0.25, and 0.125 km. The calculated fractal dimensions for every cell containing a fault were then interpolated to obtain a contour map of fractal dimensions (Figs. 4 and 5). The highest values of D (i.e., > 1.40) are distributed in the central parts of the study area, mainly corresponding to the Laochang and Gaosong Sn fields (Figs. 2 and 5).

To examine the relationship between fractal dimensions of spatial distributions of faults and Sn resources, the areas of concealed Sn orebodies were projected vertically onto the contour map of fractal dimensions of fault spatial distributions. The average (D_A) values of fractal dimensions of fault spatial distributions within the individual surface-projected orebody areas were obtained, as well as the ratios (P_o/P_T) of individual surface-projected orebody areas (P_o) to the total surface-projected orebody area (P_T). There exists a strong positive correlation between P_o/P_T and D_A (Fig. 6), which is consistent with the strong positive correlation coefficient between D and RA given in Table 2. Fig. 6 shows that, in the study area, (a) Sn orebodies at depth coincide with zones of high fractal dimensions of fault spatial distributions (i.e., > 1.40) at the surface and (b) the higher the fractal dimensions of fault spatial distributions at the surface the larger the surface-projected areas of Sn orebodies. These imply that (a) a map of the fractal dimensions can be used as a vector

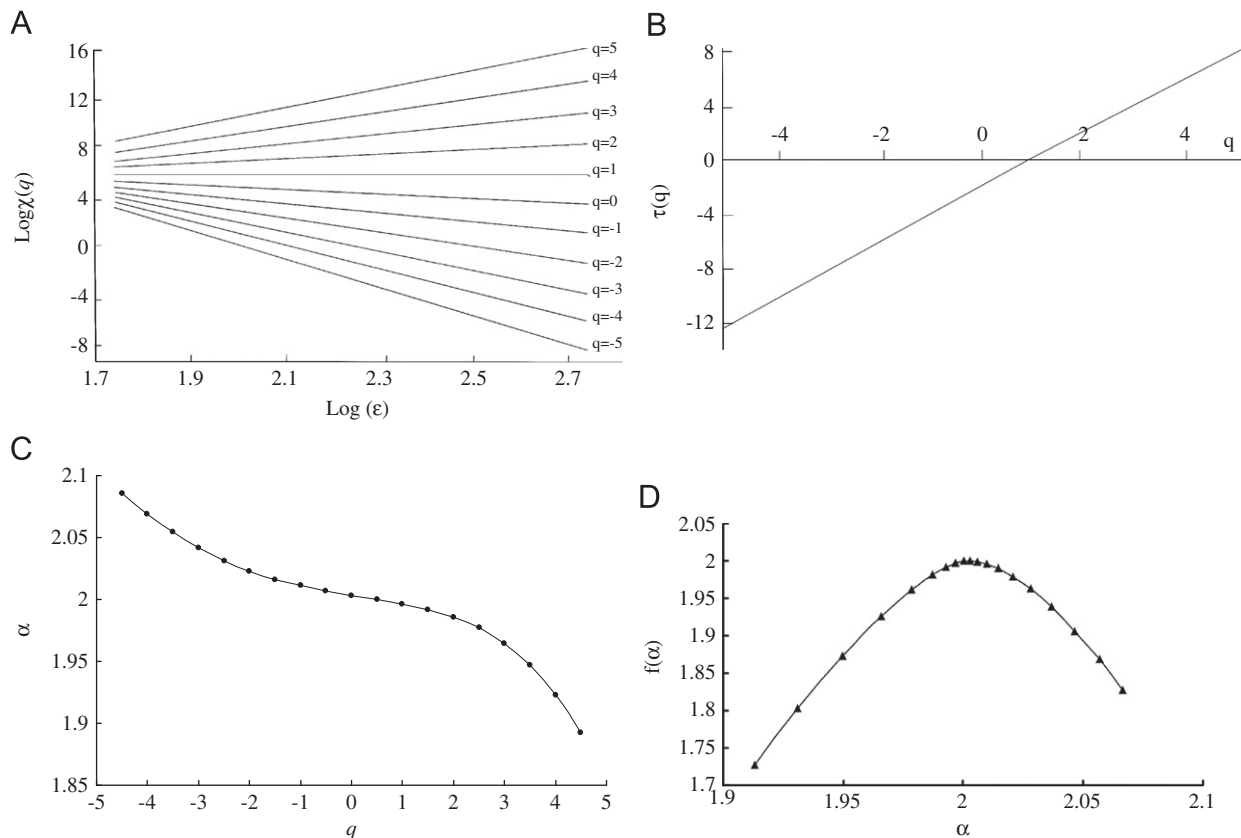


Fig. 7. Results of multifractal analysis: (A) log–log plots of $\chi_q(\epsilon)$ versus ϵ , (B) estimates of $\tau(q)$ include slopes of straight lines in A, (C) singularity $\alpha(q)$ estimated from (B) by central differentiation method and (D) multifractal spectrum $f(\alpha)$.

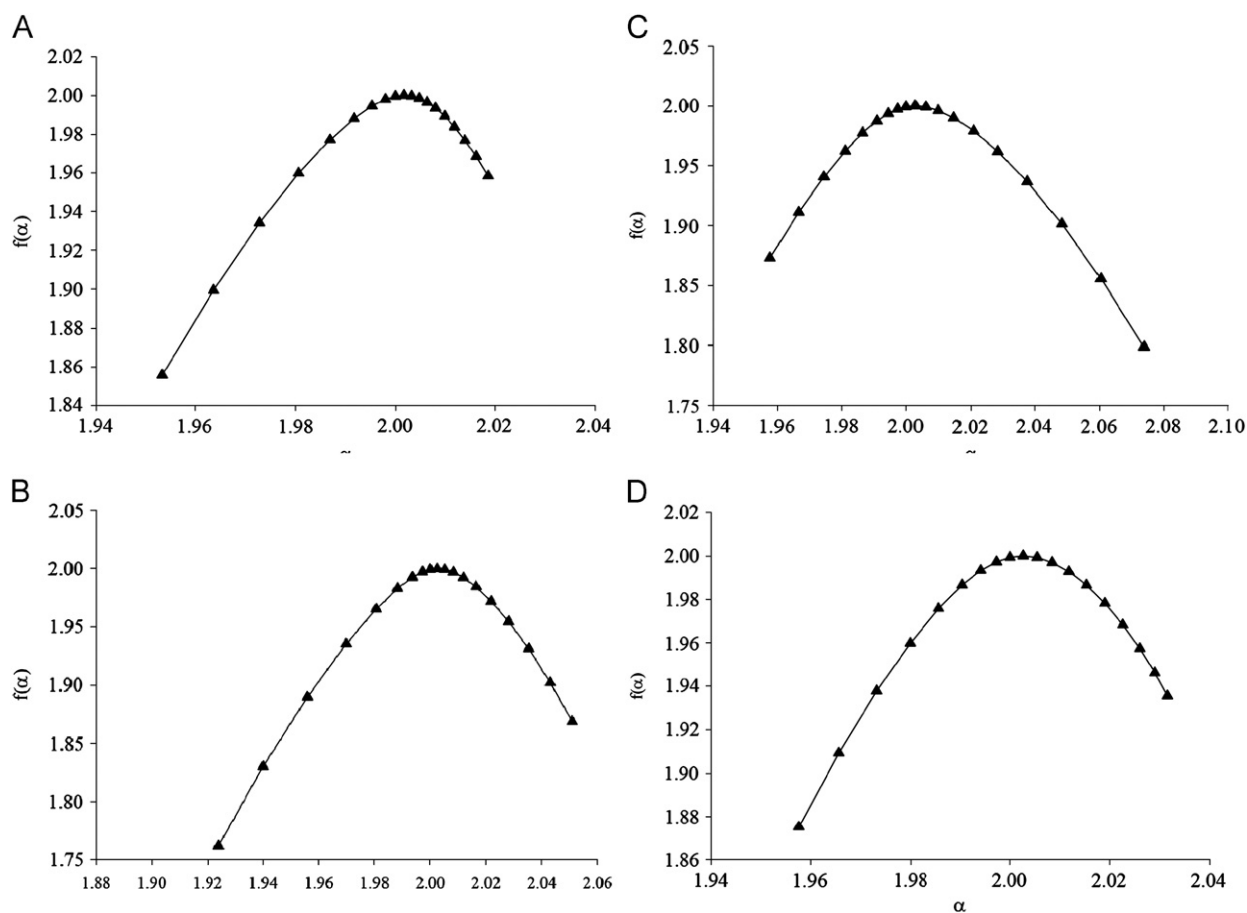


Fig. 8. Multifractal spectrum of spatial distribution of faults in each Sn field: (A) Malage; (B) Gaosong; (C) Laochang; and (D) Kafang.

toward zones where Sn orebodies likely exist and that (b) fractal dimensions of fault spatial distributions at the surface reflect indirectly the sizes of Sn orebodies at depth.

Zones of high fractal value dimensions of fault spatial distributions in the study area roughly form three NE-trending belts (Fig. 5). A review of exploration works in the study area (e.g., Zhuang et al., 1996; Chen et al., 2009) suggests that large Sn orebodies likely occur at depth along each of those NE-trending belts with a high complexity of fault patterns. The western belt (I) coincides with a granite-related skarn (Type III) Sn mineralized zone in the Malage Sn field. The middle belt (II) coincides with granite-related skarn (Type III) and stratiform (Type II) Sn mineralized zones along the axis of the Wuzishan anticlinorium (Fig. 1), which hosts the Sn deposits in the district. The eastern belt (III) coincides with a granite and gabbro-related contact skarn (Type III) Sn–Cu mineralized zone in the Laochang and Kafang Sn fields. The three NE-trending belts of high fractal dimensions of fault spatial distributions likely represent zones of upheaval related to concealed magma (Chen et al., 2009).

For the multifractal modeling of the spatial distribution of faults, a log–log plot of $\chi_q(\varepsilon)$ and ε is obtained with straight lines for small ε (Fig. 7A). Values of $\tau(q)$ were estimated from those straight lines by means of least-squares fitting with standard errors less than 0.05 and values of R^2 greater than 0.99 (Fig. 7B). The singularity exponent $\alpha(q)$ was then derived by numerically central differentiation of $\tau(q)$ (Fig. 7C). Finally, the multifractal spectrum $f(\alpha)$ is calculated by a Legendre transform (Fig. 7D).

Each multifractal spectrum for the fault patterns in each four Sn fields and for the different sets of faults in the study area (Figs. 8 and 9) appears to be arch types, continuous, and mainly

left skewed, except in the Laochang Sn field (Fig. 8C). These results suggest that the lengths of faults in each cell (i.e., in every part of the study area) exhibit multifractal properties. Two parameters are used for integration of the results. The first is $\tau''(1)$, which was defined as follows (Cheng, 1997a, 1997b, 2000): $\tau''(1) = \tau(2) - 2\tau'(1) + \tau(0)$, where $\tau'(1)$, 0 such that $\tau''(1) \approx D_2 - D_0$, where D is the information dimension. The $\tau''(1)$ is a measure of irregularity of spatial dispersion patterns, such that $\tau''(1) < 0$ suggests a multifractal spatial analysis parameter whereas $\tau''(1) = 0$ suggests an ordinary spatial analysis parameter (Cheng, 2000). The second parameter is $\Delta\alpha$, which is defined as $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$, where $\alpha_{\min}$ and $\alpha_{\max}$ are the minimum and maximum values of α , respectively. The $\Delta\alpha$ is the range of singularity index α . Values of $\Delta\alpha$ depict degrees of local irregularity and complexity characteristics.

All the obtained values of $\tau''(1)$ are less than 0, indicating that the lengths of faults in the whole study area and in each of the four Sn fields exhibit multifractal distribution patterns (Table 3). The value of $\Delta\alpha$ for all faults is greater than those for NW- and NE-trending faults, and the value of $\Delta\alpha$ for NW-trending faults is higher than for NE-trending faults. These indicate that the spatial distributions of all faults is more irregular than the spatial distributions of NE- or NW-trending faults, whereas the spatial distribution of NW-trending faults is more irregular than spatial distribution of NE-trending faults. There is a strong positive correlation among ranges of singularity index and fractal dimensions, Sn deposit attributes, fault attributes, and fractal dimensions of fault distributions in the four Sn fields in Gejiu district (Table 4). The positive correlation between the ranges of singularity index α and the surface-projected areas of Sn orebodies suggests that the former can be used as a vector toward

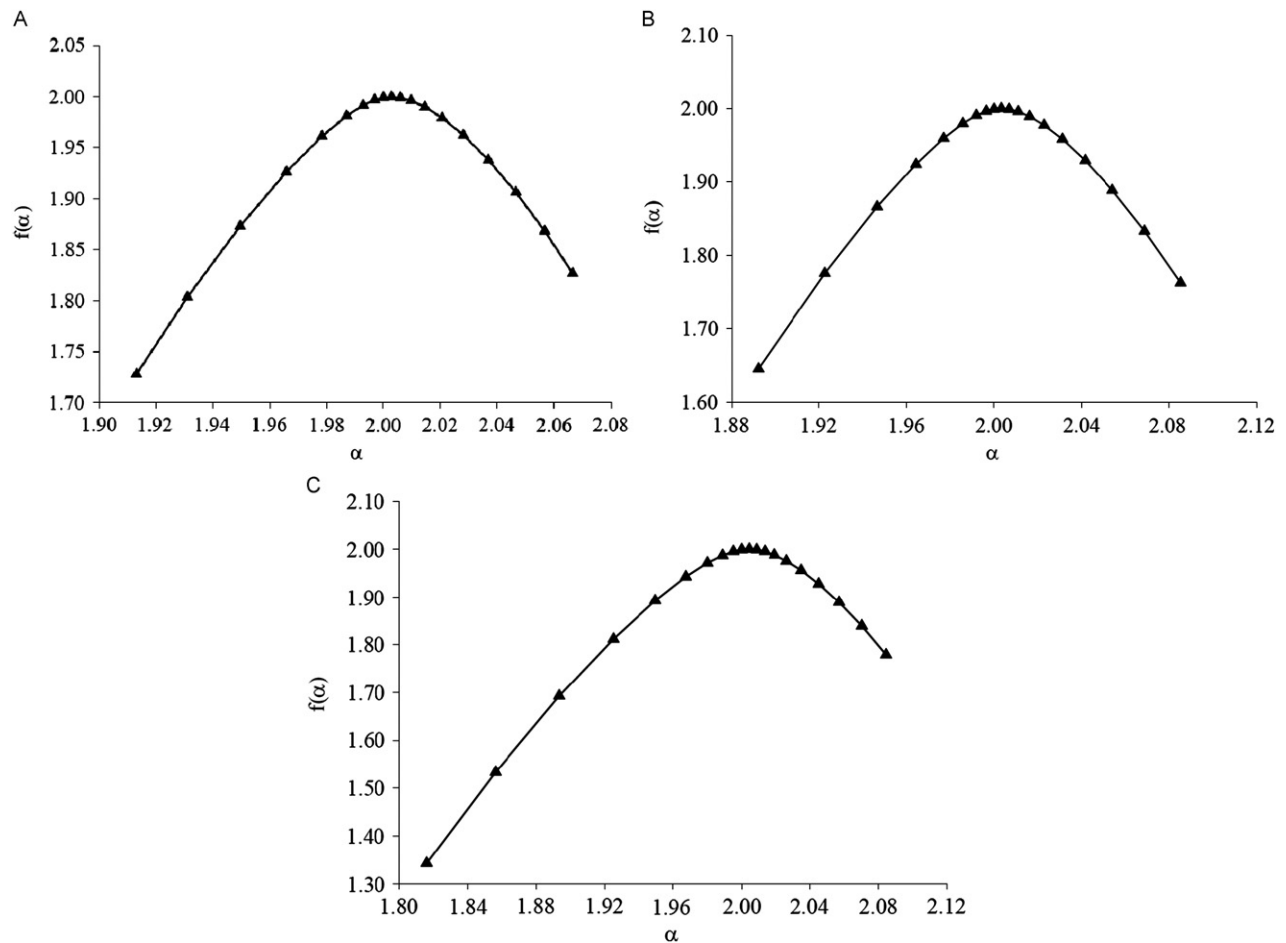


Fig. 9. Multifractal spectrum of spatial distribution of different sets of faults in the study area (Fig. 2): (A) all faults; (B) NE-trending faults; and (C) NW-trending faults.

Table 3

Multifractal properties of spatial distributions of all NE- and NW-trending faults in the study and all faults in each of the four Sn fields.

	All	NE trending	NW trending	Malage	Gaosong	Laochang	Kafang
$\tau^*(1)$	−0.0112	−0.0071	−0.0081	−0.0038	−0.0065	−0.0056	−0.0058
$\Delta\alpha$	0.228	0.153	0.193	0.061	0.125	0.126	0.073

Table 4

Pearson's correlation coefficients among range of singularity index α and fractal dimensions, Sn deposit attributes, fault attributes, and fractal dimensions of fault distributions in the four Sn fields in Gejiu district (see Tables 1 and 3).

	$\Delta\alpha$
D	0.98
FN	0.97
FL	0.96
RA	0.86
RP	0.99

Abbreviations: $\Delta\alpha$, range of singularity index α ; D, fractal dimension; FN, no. of fault segments; FL, total length of faults; RA, surface-projected areas of Sn orebodies; RP, % of Sn resources in the district.

zones in the study area where Sn ores likely occur. It also suggests that the ranges of singularity index reflect indirectly the sizes of Sn orebodies at depth.

5. Conclusions

This study focused on the fractal and multifractal properties of the spatial distribution of mapped faults in Gejiu district (China). The fractal analysis was carried out for mapped faults, and indices of the complexity of fault are displayed as a contour map. This map highlights the relationship between distribution of Sn resources and complexity of spatial distribution of faults in the Gejiu district. Zones with high complexity indices, represented by fractal dimensions, likely experienced multiple processes that played critical roles in the formation of Sn deposits in the Gejiu district. This study also showed that lengths of faults in the Gejiu area exhibit multifractal properties, and multifractal modeling is suitable for describing the complexity of the spatial distribution of faults. Although the $\Delta\alpha$ and fractal dimension have different meanings, they both represent, in the context of this study, measures of the heterogeneity of the spatial distribution of faults (cf. Cheng, 2002). That is because they are conceptually related, their values are numerically correlated (Table 4), and both of

them suggest that the complexity of the spatial distribution of faults can be used as vectors toward ore-bearing zones and they are related to sizes of orebodies in the study area.

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